Department of Electrical and Computer Engineering
COMSATS University Islamabad, Attock Campus

Lab Rubrics Form		
Lab Name	Basic Programming Applications Using Arithmetic and Conditional Statements in C	
Lab Number	05	
Submitted by:		
Student Names	**Registration #**	
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Rubrics		**Marks**		
		1	**2**	**3**
PLO-5 **Modern Tools Usage**	Ability to execute the experiment utilizing hardware, software tools, or a combination of both.			
PLO-10 **Communication Skills**	Present lab activities effectively through detailed lab reports, oral examinations, and engaging presentations.			

Lab 05: Basic Programming Applications Using Arithmetic and Conditional Statements in C

Objective

To apply arithmetic expressions, conditional statements, built-in operators, and input/output functions in C programming for solving practical engineering and mathematical problems.

Software Requirements

- Code::Blocks IDE / Dev-C++
- GCC Compiler
- Windows Operating System

Experiments Performed

In-Lab Tasks

1. Calculate the area and perimeter of a rectangle.
2. Calculate John's gross salary using basic salary, dearness allowance, and house rent allowance.
3. Display the ASCII value of a character.
4. Determine the memory size of different data types using the sizeof() operator.
5. Calculate the sum of the digits of a five-digit number.
6. Convert temperature from Centigrade to Fahrenheit.

Post-Lab Tasks

1. Calculate total expenses after applying a discount based on quantity purchased.
2. Compare two numbers and determine whether one is greater, smaller, or equal to the other.

Learning Outcomes

After completing this lab, the student will be able to:

- Perform arithmetic calculations using C operators.
- Use the sizeof() operator to determine data type sizes.
- Convert measurement units and temperatures.

- Apply conditional statements for decision-making.
- Develop programs for solving practical numerical problems.
- Strengthen understanding of variables, operators, and user input.

Task: 01

The length &breadth of rectangle are input through the keyboard. Write a program to calculate the area & perimeter of the rectangle?

Code:

```
#include <stdio.h>
#include <stdlib.h>
int main()
{
   Int Length, Width;
   Int AoR, PoR;
   Printf(“Enter the Length of Rectangle\n”);
   Scanf(“%d”,&Length);
   Printf(“Enter the Width of Rectangle\n”);
   Scanf(“%d”,&Width);
    AoR = Length*Width;
    Printf(“\t\nArea of Rectangle=%d\n”,AoR);
   PoR = 2*Length + 2*Width;
  Printf(“Parameter of Rectangle=%d\n”, PoR);
  return 0;
}
```

Output:

```
Enter the Length of Rectangle
4
Enter the Width of Rectangle
5

Area of Rectangle=20
Parameter of Rectangle=18

Process returned 0 (0x0)   execution time : 4.328 s
Press any key to continue.
```

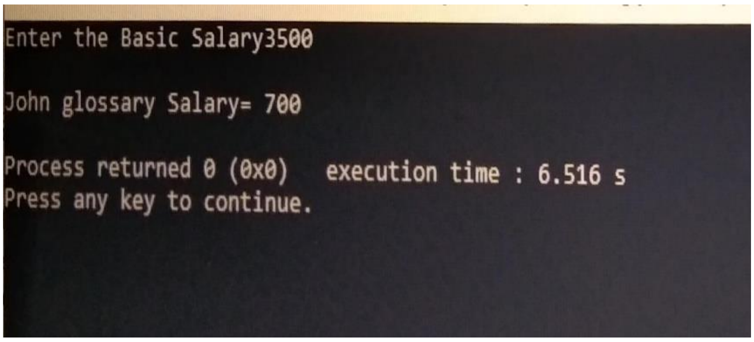
Task: 02

John's basic salary is input through the keyboard. His dearness allowance is 50% of basic salary, and house rental allowances 30% of basic salary. Write a program to calculate his gross salary.?

Code:

```
#include <stdio.h>
#include <stdlib.h>
  /* Find The John Glossary Salary*/
int main()
{
  int John_B_S, John_G_S;
  int D_A , H_r_A;
  printf("Enter the Basic Salary");
  scanf("%d", &John_B_S);
  D_A= John_B_S*50/100;
  H_r_A= John_B_S*30/100;
  John_G_S= John_B_S-(D_A+H_r_A);

   printf("\nJohn glossary Salary= %d\n", John_G_S);
   return 0;
}
```

Output:



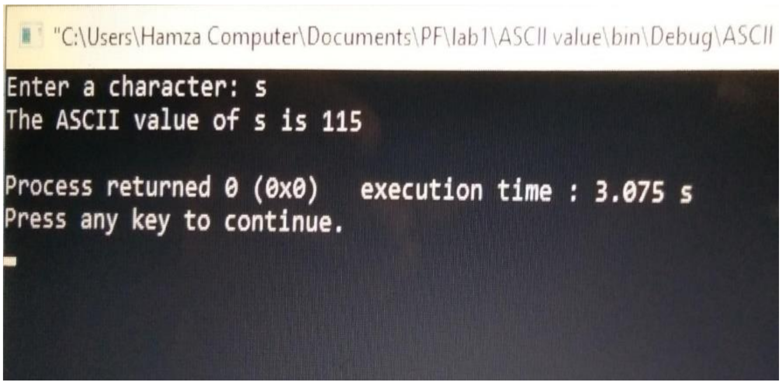
Task: 03

Write a program to print the ASCII value of a character.

Code:

```
#include <stdio.h>

#include <stdlib.h>

int main()

{

    /* ASCII value of a character */

    char c;

    printf("Enter a character: ");

    scanf("%c", &c);

    printf("The ASCII value of %c is %d\n", c, c);

    return 0;

}
```

Output:



Task: 04

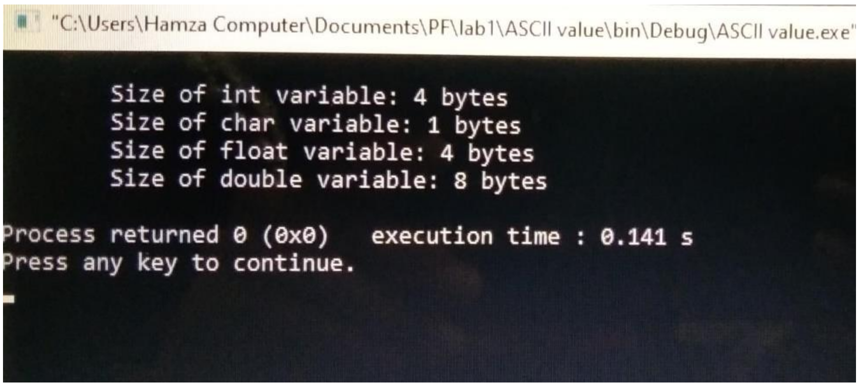
Write a program to find he size of variables having integer, char, float and double as datatype. (hint: size of (variable)).

Code:

```
#include <stdio.h>

#include <stdlib.h>
```

```
int main()
{
    int int_Type;
    char char_Type;
    float float_Type;
    double doubleType_Type;
    /*  Find the size of variables */
    printf("\n\tSize of int variable: %d bytes\n", sizeof(int_Type));
    printf("\tSize of char variable: %d bytes\n", sizeof(char_Type));
    printf("\tSize of float variable: %d bytes\n",sizeof(float_Type));
    printf("\tSize of double variable: %d bytes\n", sizeof(double_Type));
    return 0;
}
```

Output:



Task: 05

If a 5-digit number is input through the keyboard, write a program to calculate the sum of its digits. (hint: use the modulus operator%).

Code:

```
#include <stdio.h>
#include <stdlib.h>
```

```
int main()
{
    int num;
    int p,q,r,s,t;
    int pp,qq,rr,ss,tt;
    printf("Enter the five digit number:\n");
    scanf("%d", &num);
    t = num%10;
    ss = num%100;
    s = ss/10;
    rr = num%1000;
    r = rr/100;
    qq = num%10000;
    q = qq/1000;
    pp = num%100000;
    p = pp/10000;
    printf("the reverse of the number\n%d\n%d\n%d\n%d\n%d",t,s,r,q,p);
    return 0;
}
```

Task: 06

Temperature of a is input through the keyboard. Write program to convert this temperature into a Fahrenheit degrees. (T(°F)=T(°C)×1.8+32)

Code:

```
#include <stdio.h>
#include <stdlib.h>
int main()
{
    int Tempf;
    int Tempc;
    printf("\t\n Enter the Temperature in Centigrade\n",Tempc);
    scanf("%d", &Tempc);
```

```
    /* Find Temperature in Farenhiet */
    Tempf=(Tempc*1.8+32);
    printf(" Temperature in Centigrade=%d\n",Tempf);
    return 0;
}
```

Post Lab:

Task #1:

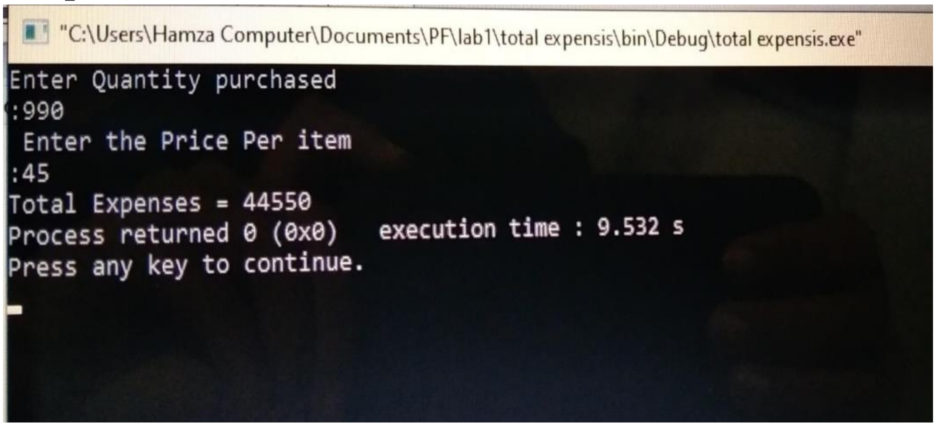
While purchasing certain items, a discount of 10%is offered if the quantity purchased is more than 1000. If quantity and price per item are input through the keyboard, write a program to calculate the total expenses.

Code:

```
 #include <stdio.h>
#include <stdlib.h>
int main()
{
    int Quantity, Price_per_item;
    int Total_Expenses;
    printf("Enter Quantity purchased\n:");
    scanf("%d", & Quantity);
    printf(" Enter the Price Per item\n:");
    scanf("%d",&Price_per_item);
    Total_Expenses = Quantity* Price_per_item;
    if(Quantity>1000)
{
     Total_Expenses = Total_Expenses *10 /100;
    printf("Total Expenses after applying discount = %d",Total_Expenses);
}
```

```
    else
        printf("Total Expenses = %d",Total_Expenses);
    return 0;
}
```

Output:



Task: 02

Take two numbers from the users as input. Check whether the one number is greater, smaller or equal to the other number.

Code:

```
#include <stdio.h>
#include <stdlib.h>
int main()
{
    int k,l;
    printf(" Enter the first Number:\n");
    scanf("%d", &k);
    printf(" Enter the 2nd Number:\n");
    scanf("%d", &l);
    if (k>l)
        printf(" First Number is Greater to the 2nd Number\n");
    else
```

printf(" First Number is Smaller or Equal to the 2nd Number\n");

Output: